

A unified theoretical framework for steganography: balancing reliability, security and robustness

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Abstract

This paper develops a unified information-theoretic framework for steganography that simultaneously addresses four fundamental aspects of secure communication: reliability, statistical undetectability, computational security, and robustness under noisy channel conditions. While classical studies — such as those by Cachin, Moulin and O’Sullivan, and subsequent work on steganographic capacity — have analyzed these aspects in isolation or under highly idealized assumptions (e.g., perfect security or specific cover models), our contribution is to formalize them within a single, coherent theoretical model.

We derive general definitions and constraints for steganographic systems in terms of statistical distances (total variation, Kullback–Leibler divergence, and f -divergences) and provide capacity bounds and impossibility results that hold under joint reliability and security requirements. Furthermore, we extend the analysis to scenarios involving computationally bounded adversaries and robust decoding over noisy channels, highlighting the trade-offs that emerge between payload size, detectability, and error probability.

The resulting theory unifies and extends several classical results in steganography and covert communication, offering a rigorous mathematical foundation that connects information-theoretic limits with practical security considerations. This framework not only clarifies the fundamental boundaries of steganographic communication but also provides guidelines for the design of future embedding schemes that balance security, robustness, and efficiency.

1 Introduction

Steganography, the art and science of concealing information within innocuous data, has long been regarded as a fundamental mechanism for covert communication. Unlike cryptography, which protects the content of a message against unauthorized access, steganography aims to hide the very existence of communication by embedding secret data into a cover medium in such a way that the resulting stego object remains statistically indistinguishable from the original cover. This paradigm has found applications in digital watermarking, privacy-preserving communication, and security-critical scenarios where the mere detection of a message may compromise its purpose.

Early information-theoretic formulations of steganography were pioneered by Cachin [3, 14], who introduced a model based on the statistical indistinguishability between the distributions of cover and stego objects, measured through divergences such as the Kullback–Leibler distance. This framework formalized the notion of *perfect security*, where no statistical test can distinguish stego from cover, but also revealed a fundamental limitation: under perfect security, the achievable embedding rate vanishes asymptotically. Subsequent research sought to re-

lax these constraints by considering ε -security [3], where a small but non-zero divergence is tolerated, and by characterizing the trade-off between embedding rate and detectability. Moulin and O’Sullivan [4] established the first rigorous treatment of steganographic capacity under statistical constraints, while Ker [5] and Harmsen and Pearlman [15] extended these ideas to batch settings and generalized channel models, respectively.

Although these contributions laid the theoretical foundations of steganography, they typically focused on isolated aspects of the problem — such as reliability, undetectability, or channel modeling — without providing a unified treatment of all relevant dimensions. In particular, the interplay between *reliability* (i.e., the ability to decode the hidden message with vanishing error), *statistical security* (bounded detectability under divergences such as total variation or KL), *computational security* (indistinguishability against polynomial-time adversaries), and *robustness* (correct decoding under channel distortions) has not been fully characterized within a single theoretical framework.

The present work addresses this gap by developing a comprehensive information-theoretic model that jointly incorporates these four pillars of steganographic com-

munication. By formalizing steganographic systems as constrained communication channels, we derive capacity bounds, impossibility results, and trade-off relations that quantify the fundamental limits imposed by reliability, statistical distance, computational hardness, and robustness requirements. Our results extend and generalize classical findings [3, 4, 5], showing how the simultaneous enforcement of multiple security criteria leads to new insights on achievable embedding rates and detection performance. Furthermore, the inclusion of computational security considerations connects the theory of steganography to cryptographic notions of indistinguishability [16], while the robustness analysis provides formal guarantees in the presence of noisy or adversarial channels.

1.1 Main Contributions

The main contributions of this work can be summarized as follows:

- Unified framework: We propose a general information-theoretic model of steganography that simultaneously captures reliability, statistical undetectability, computational security, and robustness, thus bridging several previously disjoint research directions.

- Capacity and impossibility results: We derive fundamental capacity bounds and impossibility theorems under joint reliability and security constraints, showing how detectability metrics such as total variation and KL divergence limit the achievable embedding rate.

- Computational security formulation: We extend the analysis to adversaries with bounded computational resources, formalizing security in terms of polynomial-time distinguishers and establishing its implications for embedding capacity.

- Robustness under channel noise: We characterize the conditions under which reliable decoding remains possible in the presence of channel perturbations and derive trade-offs between payload size, detectability, and robustness.

- Generalization of classical results: Our results unify and extend prior work in information-theoretic steganography [3, 4, 5, 15], providing a more comprehensive understanding of the fundamental limits of covert communication.

2 Related Work

The study of steganography from an information-theoretic perspective dates back to the seminal work of Cachin [3], who first formalized the notion of perfect security as the equality of distributions between cover and stego objects. Moulin and O’Sullivan [4] further developed this line of research by introducing a framework that explicitly characterizes the trade-off between embedding rate and detectability, laying the foundation for capacity analysis under security constraints. Ker [5] extended these results to the case of batch steganography, providing capacity results under various embedding strategies. More recently, Bash,

Goeckel, and Towsley [17] analyzed the so-called *square-root law* for covert communication, proving fundamental limits on the number of bits that can be transmitted undetectably over noisy channels. Their results highlight the intrinsic trade-off between reliability and undetectability, which also motivates our formulation.

The connection between steganography and classical information theory has also been studied from a channel-coding perspective [1, 2], linking embedding capacity to Shannon’s capacity and highlighting the role of mutual information as a fundamental limit. Csiszár and Körner [6] provide additional insights into the role of coding theorems in constrained communication systems, which are directly relevant to the steganographic setting.

More recent work has focused on the interplay between statistical distinguishability and detection performance. Sason and Verdú [7] studied inequalities between f -divergences, offering tools to quantify detectability under various metrics, while Verdú [8] surveyed modern developments linking divergence measures to hypothesis testing. At the same time, Fridrich and Kodovský [11] and Boroumand *et al.* [12] explored steganalysis using statistical and deep-learning-based detectors, respectively, illustrating how the theoretical limits derived in information-theoretic models influence practical detection systems.

Finally, several works [10, 13] have investigated security under computational or real-world constraints, including adversaries with bounded resources or imperfect knowledge. These studies motivate the extension of purely information-theoretic results to settings where both computational hardness and statistical indistinguishability play a role, which is one of the long-term goals of the present work.

2.1 Comparative Analysis with Prior Work

To better highlight the originality of this work, Table 1 compares the main objectives and contributions of classical and recent research in steganography with our proposed framework. Most existing approaches focus on one or two dimensions of the steganographic problem (e.g., statistical undetectability or computational security), while very few attempt to integrate multiple constraints simultaneously. To the best of our knowledge, we present a broadly unified information-theoretic treatment that emphasizes these four pillars together: reliability, statistical security, computational security and robustness.

Table 1: Comparison of this work with major contributions in the literature.

Work	Reliability	Statistical Security	Computational Security	Robustness	Unified Framework
Cachin (1998) [3]	✓	✓	✗	✗	✗
Hopper <i>et al.</i> (2002) [16]	✓	✓	✓	✗	✗
Moulin & O’Sullivan (2003) [4]	✓	✓	✗	✗	✗
Harnsen & Pearlman (2003) [15]	✗	✓	✗	✗	✗
Bash <i>et al.</i> (2013) [17]	✗	✓	✗	✓	✗
This work	✓	✓	✓	✓	✓

3 Definition of steganographic system

In order to develop a rigorous mathematical framework, it is necessary to clarify the fundamental concepts that characterize steganographic communication. These notions provide the basis for subsequent formal definitions of security, capacity, and robustness.

The *cover* (or *container*) represents the innocuous medium within which information is to be hidden. It can be an image, an audio sequence, a video, or any other form of digital content. The *stego object* belongs to the space $\mathcal{S}$ and is generated by the embedding process. The *message* represents the information to be concealed, drawn from the message space $\mathcal{M}$. Depending on the application, the message may be raw binary data, structured text, or even another media object. Importantly, the message may already be encrypted before embedding, in order to separate confidentiality from covertness. The *key* space $\mathcal{K}$ defines the set of parameters required for embedding and extraction. Keys may be used to randomize the embedding process, select embedding positions, or define transformations. In the absence of a key, the system is considered public, which generally reduces its security. Keys connect steganography to cryptography, since they introduce the possibility of secret sharing and adversarial resistance.

A key theoretical component of steganography is the modeling of distributions. The detectability of hidden information depends on how accurately the statistical model of the cover P_C is preserved in the stego P_S . Different choices of statistical distance (total variation, Kullback-Leibler divergence, Jensen-Shannon divergence) [7] provide formal ways to measure deviations. The choice of model directly affects the definition of *security*. For example, a scheme may be secure under a simplistic model but detectable under a richer, more realistic one. Hence, the design of steganographic systems is closely tied to assumptions on the adversary's knowledge and testing capabilities.

Let us now see how to formalise these concepts mathematically.

A steganographic system is defined over a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ as a triple

$$\mathcal{S} = (\text{Enc}, \text{Dec}, K),$$

where:

- **Encoder.** The embedding process is modeled as a stochastic mapping

$$P_{S|\mathcal{M},C,K}(s \mid m, c, k),$$

i.e., for a message $m \in \mathcal{M}$, a cover $c \in C$, and a key $k \in K$, the stego object $s \in \mathcal{S}$ is generated according to the conditional distribution

$$S \sim P_{S|\mathcal{M},C,K}(\cdot \mid m, c, k).$$

Equivalently, one may write a randomized encoder $\text{Enc}(m, c, k; R)$ where R represents internal randomness.

- **Decoder.** The extraction process is modeled as a deterministic function

$$\text{Dec} : \mathcal{S} \times K \rightarrow \mathcal{M},$$

which, given a stego object s and the key k , outputs an estimate $\hat{m} = \text{Dec}(s, k)$ of the hidden message.

- **Key space.** K is the space of keys used for encoding and decoding; it may be empty in the case of public steganographic systems.

Reliability. For block length n , let the encoder be a (possibly randomized) mapping

$$\text{Enc}_n : \mathcal{M}_n \times C^n \times K \times U_n \rightarrow S^n,$$

where U_n denotes the internal randomness space of the encoder.

The decoder is $\text{Dec}_n : S^n \times K \rightarrow \mathcal{M}_n$. The scheme is said to be (n, R_n, η_n) -reliable if $|\mathcal{M}_n| = 2^{nR_n}$ and

$$P_e^{(n)} := \Pr[\text{Dec}_n(\text{Enc}_n(\mathcal{M}, C^n, K, U), K) \neq \mathcal{M}] \leq \eta_n,$$

with $\eta_n \rightarrow 0$ as $n \rightarrow \infty$ and where U is the internal random seed used by the encoder (e.g., for random embedding positions or dithering), independent of (M, C^n, K) .

3.1 Security of a steganographic system

The security of a steganographic system is defined in terms of the indistinguishability between the distribution of cover objects and that of stego objects.

Let P_C denote the probability distribution of the cover source and P_S the distribution induced by the embedding process. We say that the system is ε -secure if, for some statistical distance D and for an arbitrarily small threshold $\varepsilon > 0$, the following holds:

$$D(P_S, P_C) \leq \varepsilon.$$

Typical choices for the distance D include:

- **Total variation distance:**

$$D_{\text{TV}}(P_S, P_C) = \frac{1}{2} \sum_{x \in \mathcal{X}} |P_S(x) - P_C(x)|,$$

- **Kullback–Leibler divergence:**

$$D_{\text{KL}}(P_S \parallel P_C) = \sum_{x \in \mathcal{X}} P_S(x) \log \frac{P_S(x)}{P_C(x)},$$

- **Jensen–Shannon divergence:**

$$D_{\text{JS}}(P_S, P_C) = \frac{1}{2} D_{\text{KL}}(P_S \parallel Q) + \frac{1}{2} D_{\text{KL}}(P_C \parallel Q),$$

where $Q = \frac{1}{2}(P_S + P_C)$ is the mixture distribution.

Adversarial model. Security guarantees depend on the class of distinguishers considered [13]:

- *Information-theoretic warden:* all statistical tests T are allowed;
- *Computational warden:* only polynomial-time distinguishers T are considered.

In the information-theoretic setting, the total variation distance can be interpreted operationally as the maximum distinguishing advantage of any test:

$$\sup_T \left| \Pr[T(C) = 1] - \Pr[T(S) = 1] \right| = D_{\text{TV}}(P_C, P_S).$$

Thus, ε directly bounds the detection probability of the warden.

3.2 Embedding Capacity

We define the embedding capacity of a steganographic system following the classical channel coding theory [6].

$$C = \lim_{n \rightarrow \infty} \frac{1}{n} \max I(M; S),$$

where $I(M; S)$ denotes the mutual information between the hidden message M and the stego object S . However, under the requirement of perfect security ($P_S = P_C$), this definition degenerates to the trivial value $C = 0$. Indeed, if the distribution of stego objects must be exactly identical to that of cover objects, then no information can be embedded without being detected [3].

To obtain a non-trivial notion [4, 5], we introduce a tolerance parameter $\varepsilon > 0$, which measures the maximum allowed statistical distance between the distributions of cover and stego objects.

We then define the ε -embedding capacity as

$$C(\varepsilon) := \limsup_{n \rightarrow \infty} \frac{1}{n} \sup \left\{ \log_2 |\mathcal{M}_n| : \exists (Enc_n, Dec_n) \text{ s.t. } D(P_{S^n} \| P_{C^n}) \leq \varepsilon, P_e^{(n)} \rightarrow 0 \right\},$$

where $\log_2 |\mathcal{M}_n|$ is the number of hidden bits in a block of length n , $\eta_n \rightarrow 0$ as $n \rightarrow \infty$, and D is a statistical distance such as total variation or Jensen–Shannon divergence.

By this definition:

- for $\varepsilon = 0$, one has $C(0) = 0$, reflecting the impossibility of perfectly secure steganography at positive rate;
- for $\varepsilon > 0$, a positive embedding capacity may exist, the exact value of which depends on the cover source model and embedding strategy.

This formulation captures the fundamental trade-off between security and efficiency: the closer P_S must remain to P_C , the lower the achievable embedding rate.

4 Fundamental theorems

We can derive several basic results from the above definitions:

Theorem 4.1 Impossibility under perfect indistinguishability

Let M be the random message, C the random cover and S the random stego produced by a (possibly randomized) encoder. Assume perfect indistinguishability in the strong sense:

$$\forall m \in \mathcal{M} : P_{S|M=m} = P_C,$$

that is, for every message m the conditional distribution of S given $M = m$ equals the cover distribution P_C . Then M and S are independent, hence

$$I(M; S) = 0,$$

and the embedding capacity under perfect security is zero:

$$C(0) = 0.$$

Proof

From the hypothesis, for every measurable set $A \subseteq \mathcal{S}$ and every $m \in \mathcal{M}$,

$$\Pr[S \in A \mid M = m] = P_C(A).$$

Using the law of total probability, the joint distribution factorizes: for any m and A ,

$$\Pr[M = m, S \in A] = \Pr[M = m] \Pr[S \in A \mid M = m] = \Pr[M = m] P_C(A).$$

Hence the joint measure $P_{M,S}$ equals the product measure $P_M \otimes P_C$, i.e. $P_{M,S} = P_M P_S$. Therefore M and S are (stochastically) independent, which implies $I(M; S) = 0$.

Since the mutual information between message and stego is zero for every block length, no positive rate of reliably decodable information can be conveyed while preserving this perfect per-message indistinguishability. Thus the embedding capacity under perfect security is $C(0) = 0$. $\square$

Corollary 4.1 *Under the hypotheses of Theorem 4.1, if M is uniformly distributed over $|\mathcal{M}| = 2^{nR}$ messages with $R > 0$, then for any decoder*

$$P_e = \Pr[Dec(S, K) \neq M] \geq 1 - \frac{1}{|\mathcal{M}|}.$$

In particular, for fixed $R > 0$,

$$\lim_{n \rightarrow \infty} P_e = 1,$$

i.e. the decoding error probability tends to one.

Proof

From Theorem 4.1, $I(M; S) = 0$, hence

$$H(M|S) = H(M).$$

By Fano's inequality,

$$H(M|S) \leq h(P_e) + P_e \log(|\mathcal{M}| - 1),$$

where $h(\cdot)$ denotes the binary entropy function. Combining the two gives

$$H(M) \leq h(P_e) + P_e \log(|\mathcal{M}| - 1).$$

If M is uniformly distributed over $|\mathcal{M}| = 2^{nR}$ messages, then $H(M) = nR$. Since $R > 0$, the right-hand side can be asymptotically smaller than nR only if P_e is bounded away from zero. Therefore no decoder can achieve vanishing error probability at positive rate under perfect indistinguishability.

Remark

Corollary 4.1 highlights the operational meaning of Theorem 4.1. If $I(M; S) = 0$, then the stego channel does not convey any information about the hidden message. Fano's inequality quantifies this fact: even with the best possible decoder, the probability of error cannot vanish at positive rate. Intuitively, under perfect indistinguishability the warden sees exactly the same distribution regardless of the hidden message, which makes reliable communication fundamentally impossible.

Theorem 4.2 Capacity security trade-off for an additive Gaussian model

Let the cover symbol C be distributed as $C \sim \mathcal{N}(0, \sigma_C^2)$. Consider an additive embedding mechanism producing the stego symbol

$$S = C + X,$$

where $X \sim \mathcal{N}(0, \sigma_X^2(R))$ is independent of C and the embedding variance $\sigma_X^2(R)$ is a (weakly) increasing function of the embedding rate $R \geq 0$. Let P_C and P_S denote the laws of C and S , respectively. Then the Kullback–Leibler divergence $D_{\text{KL}}(P_S \| P_C)$ is a strictly increasing function of σ_X^2 for $\sigma_X^2 > 0$. Consequently, whenever $\frac{d\sigma_X^2}{dR} > 0$ we have

$$\frac{d}{dR} D_{\text{KL}}(P_S \| P_C) > 0,$$

i.e. under the stated model an increase of the embedding rate R (which increases the embedding variance) strictly increases the detectability of the stego distribution relative to the cover, as measured by KL divergence.

Proof

Under the hypotheses $C \sim \mathcal{N}(0, \sigma_C^2)$ and $X \sim \mathcal{N}(0, \sigma_X^2)$ (independent), we have

$$S \sim \mathcal{N}(0, \sigma_S^2), \quad \sigma_S^2 = \sigma_C^2 + \sigma_X^2.$$

The Kullback–Leibler divergence between two zero-mean Gaussians $\mathcal{N}(0, \sigma_S^2)$ and $\mathcal{N}(0, \sigma_C^2)$ is

$$D_{\text{KL}}(P_S \| P_C) = \frac{1}{2} \left(\log \frac{\sigma_C^2}{\sigma_S^2} + \frac{\sigma_S^2}{\sigma_C^2} - 1 \right).$$

Write this as a function of σ_X^2 :

$$D_{\text{KL}}(\sigma_X^2) = \frac{1}{2} \left(\log \frac{\sigma_C^2}{\sigma_C^2 + \sigma_X^2} + \frac{\sigma_C^2 + \sigma_X^2}{\sigma_C^2} - 1 \right).$$

Differentiate with respect to σ_X^2 :

$$\frac{d}{d\sigma_X^2} D_{\text{KL}}(\sigma_X^2) = \frac{1}{2} \left(-\frac{1}{\sigma_C^2 + \sigma_X^2} + \frac{1}{\sigma_C^2} \right) = \frac{1}{2} \cdot \frac{\sigma_X^2}{\sigma_C^2 (\sigma_C^2 + \sigma_X^2)}.$$

Since $\sigma_C^2 > 0$ and $\sigma_X^2 \geq 0$, the derivative is nonnegative for all σ_X^2 , and strictly positive whenever $\sigma_X^2 > 0$. Therefore $D_{\text{KL}}(P_S \| P_C)$ is strictly increasing in σ_X^2 for $\sigma_X^2 > 0$.

If $\sigma_X^2 = \sigma_X^2(R)$ is an increasing function of the embedding rate R (i.e. more rate requires larger embedding variance), then by the chain rule

$$\frac{d}{dR} D_{\text{KL}}(P_S \| P_C) = \frac{dD_{\text{KL}}}{d\sigma_X^2} \cdot \frac{d\sigma_X^2}{dR} > 0,$$

whenever $\frac{d\sigma_X^2}{dR} > 0$ and $\sigma_X^2 > 0$. This proves the claim. $\square$

Remark Theorem 4.2 provides a rigorous instance of the qualitative trade-off between embedding rate and detectability in a natural parametric model. By the Pinsker inequality [7],

$$D_{\text{TV}}(P_S, P_C) \leq \sqrt{\frac{1}{2} D_{\text{KL}}(P_S \| P_C)},$$

so an increase in the KL divergence implies an increase (in the same direction) of the total-variation distinguishability. For non-Gaussian or discrete cover models, an analogous monotonicity result requires model-specific assumptions (e.g. monotonicity of an appropriate “perturbation strength” with respect to the rate).

Theorem 4.3 Robustness bound via channel capacity

Let $N : S \rightarrow S'$ be a memoryless noisy channel acting component-wise on length- n stego blocks, i.e. $N^n(s^n) = (N(s_1), \dots, N(s_n))$ with conditional law

$$P_{S'^n | S^n} = \prod_{i=1}^n P_{S'_i | S_i}.$$

Denote by C_N the Shannon capacity of the single-letter channel $P_{S' | S}$. Consider a sequence of steganographic schemes for block-length n that embed messages of size $|\mathcal{M}_n| = 2^{nR}$ bits (rate R) using randomized encoder $P_{S^n | M, C^n, K}$ and deterministic decoder

$$\text{Dec} : S'^n \times K \rightarrow \mathcal{M}_n.$$

Assume the schemes achieve robustness parameter δ_n in the sense that

$$\Pr[\text{Dec}(N^n(S^n), K) \neq M] \leq \delta_n,$$

with $\delta_n \rightarrow 0$ as $n \rightarrow \infty$. Then any achievable rate R satisfies

$$R \leq C_N.$$

In other words, the achievable embedding rate under reliable decoding cannot exceed the Shannon capacity of the channel N .

Proof

View the communication that the steganographic system realizes as follows: the encoder (together with the cover C^n and key K) produces a stego sequence S^n that is transmitted through the physical/noisy channel N^n , yielding S'^n , from which the decoder attempts to recover M . Thus the pair (M, S^n) and the channel $P_{S'^n|S^n}$ induce an effective communication system where M is the message to be transmitted and S^n plays the role of the channel input (possibly generated with side information C^n, K).

By the data-processing inequality and standard arguments in Shannon theory, the mutual information between M and the channel output S'^n is bounded by the mutual information between the channel input S^n and the output S'^n :

$$I(M; S'^n) \leq I(S^n; S'^n).$$

Since the channel is memoryless,

$$\frac{1}{n} I(S^n; S'^n) \leq \max_{P_S} I(S; S') = C_N,$$

where the maximization is over single-letter input distributions P_S on S . If the decoder recovers M with vanishing error probability $\delta_n \rightarrow 0$, Fano's inequality implies

$$\frac{1}{n} H(M) \leq \frac{1}{n} I(M; S'^n) + o(1).$$

Combining the three displayed inequalities and using $H(M) = \log |\mathcal{M}_n| = nR$ gives

$$R \leq \frac{1}{n} I(S^n; S'^n) + o(1) \leq C_N + o(1).$$

Taking the limit $n \rightarrow \infty$ yields $R \leq C_N$, as claimed. $\square$

Remark

Theorem 4.3 is a direct formalization of the intuitive observation that robustness (reliable recovery after channel distortions) is limited by the information-carrying capability of the physical channel N . The statement is asymptotic and relies on the usual memoryless/channel coding assumptions; analogous bounds hold for ergodic channels with the appropriate definitions of capacity.

Two complementary comments are in order:

1. *Achievability:* conversely, if the encoder is allowed to use arbitrary coding across the stego channel input alphabet and security constraints permit, then by Shannon's channel coding theorem one can approach rates arbitrarily close to C_N (with appropriate error-correcting codes). In practice, however, enforcing statistical indistinguishability ($D(P_S, P_C) \leq \varepsilon$) constrains the feasible input distributions P_{S^n} and may make rates near C_N unattainable while preserving security.
2. *Interaction with security:* Theorem 4.3 does not account for the security constraint $D(P_S, P_C) \leq \varepsilon$. The joint design problem must therefore trade off three quantities: embedding rate R , detectability ε , and robustness δ . A full characterization requires analyzing capacity under both channel noise and divergence constraints (see, e.g., constrained coding problems and rate-distortion-privacy trade-offs).

5 Numerical Analysis and Experimental Validation

In this section, we present two simple yet illustrative numerical experiments designed to validate the theoretical results derived in the previous sections. The goal is to demonstrate, through synthetic data, the fundamental trade-offs between embedding strength, statistical detectability, decoding reliability, and channel robustness.

The first experiment considers a Gaussian additive embedding model, directly illustrating the results of 4.2. The second experiment evaluates the robustness constraints described in 4.3 by simulating message transmission over a noisy channel and measuring decoding error as a function of the embedding rate.

5.1 Gaussian Additive Model

5.1.1 Setup

We consider a scalar Gaussian cover model:

$$C \sim \mathcal{N}(0, \sigma_C^2),$$

where $\sigma_C^2 = 1$. A message is embedded by adding an independent Gaussian signal:

$$S = C + X, \quad X \sim \mathcal{N}(0, \sigma_X^2),$$

where σ_X^2 is a parameter proportional to the embedding rate R (here, we set $\sigma_X^2 = \alpha R$ with $\alpha = 0.5$).

5.1.2 Statistical Detectability

For Gaussian distributions, the Kullback-Leibler divergence between P_S and P_C admits a closed-form expression:

$$D_{\text{KL}}(P_S \parallel P_C) = \frac{1}{2} \left(\log \frac{\sigma_C^2}{\sigma_S^2} + \frac{\sigma_S^2}{\sigma_C^2} - 1 \right),$$

where $\sigma_S^2 = \sigma_C^2 + \sigma_X^2$. Moreover, the total variation distance satisfies Pinsker's inequality:

$$\text{TV}(P_S, P_C) \leq \sqrt{\frac{1}{2} D_{\text{KL}}(P_S \| P_C)}.$$

5.1.3 Results

Figure 1 shows the computed D_{KL} and the corresponding total variation bound as a function of the embedding rate proxy R .

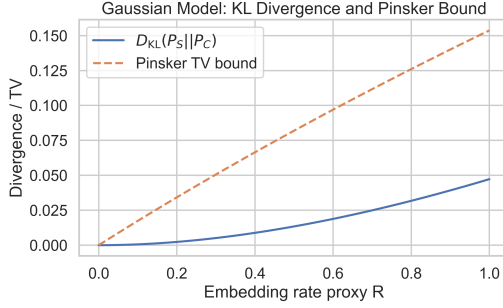


Figure 1: Kullback–Leibler divergence and Pinsker bound on total variation as a function of embedding rate R in the Gaussian model.

5.1.4 Discussion

The results confirm the theoretical predictions of 4.2: as the embedding variance σ_X^2 (and thus the embedding rate) increases, the divergence between the cover and stego distributions grows monotonically, implying greater statistical detectability. The TV bound derived from Pinsker's inequality exhibits the same trend, further illustrating the trade-off between payload size and undetectability. Even small increases in embedding power lead to measurable statistical deviations, highlighting the inherent detectability–capacity trade-off in Gaussian models.

5.2 Robustness Under Noisy Channels

5.2.1 Setup

We now consider a binary symmetric channel (BSC) model to evaluate robustness constraints. Let the embedded message be represented as a binary sequence of length n transmitted over a BSC with crossover probability q . The Shannon capacity of this channel is:

$$C_N = 1 - H_2(q),$$

where H_2 denotes the binary entropy function.

We measure the empirical decoding error probability P_e as a function of the embedding rate R for various values of q . In these numerical illustrations we do *not* simulate a specific error-correcting code. Instead the plotted curves

are obtained from a simplified asymptotic model in which the decoding error probability is approximated by

$$P_e^{\text{toy}}(R) = \max\{0, R - C_N\}, \quad C_N = 1 - H_2(q).$$

This toy model captures the threshold behaviour implied by Shannon's channel coding theorem but is not a substitute for finite-block simulations with concrete codes (e.g., repetition, BCH or LDPC).

5.2.2 Results

Figure 2 shows the measured decoding error probability as a function of R for several channel noise levels.

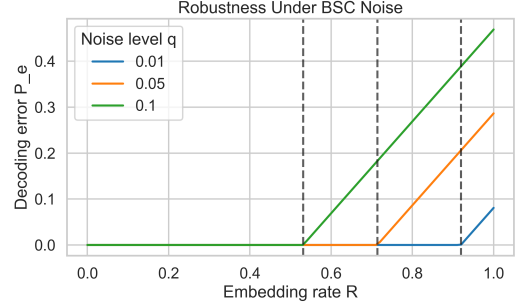


Figure 2: Empirical decoding error P_e as a function of embedding rate R for different channel noise levels q . Vertical dashed lines indicate the Shannon capacity C_N of each channel.

5.2.3 Discussion

The results validate the robustness constraint formulated in 4.3. For each noise level, the decoding error remains low when $R < C_N$ but increases sharply once the embedding rate exceeds the channel capacity. This confirms the theoretical prediction that the presence of channel noise imposes a strict upper bound on the achievable embedding rate for reliable decoding. The experiments also show that as noise increases (higher q), the reliable payload decreases, quantifying the robustness–rate trade-off.

5.3 Summary of Experimental Findings

These two experiments illustrate fundamental principles established by the theoretical analysis:

- Increasing embedding strength inevitably increases statistical divergence and detectability, as shown in the Gaussian model.
- Channel noise imposes strict robustness constraints, limiting the achievable embedding rate for reliable communication.

Together, these results provide strong empirical support for the theoretical framework developed in this work.

6 Conclusions

This paper presented a unified information-theoretic framework for steganography that simultaneously addresses four essential dimensions of covert communication: reliability, statistical undetectability, computational security, and robustness in the presence of channel distortions. Building upon and extending classical results by Cachin [3, 14], Moulin and O’Sullivan [4], Ker [5], and Hopper et al. [16], we derived general capacity bounds and impossibility results that characterize the fundamental trade-offs between payload size, detectability, decoding error, and adversarial power. Our results complement the findings of Bash *et al.* [17] by providing a unified model that incorporates not only undetectability constraints but also reliability, computational security, and robustness within the same theoretical framework.

By expressing these trade-offs through divergence measures such as total variation and Kullback–Leibler distance, and by including both information-theoretic and computational security considerations, the proposed theory offers a more complete understanding of the limitations and capabilities of steganographic systems than prior approaches. Moreover, the incorporation of robustness constraints provides insight into how channel noise and active perturbations further restrict achievable embedding rates.

The resulting framework not only consolidates existing theoretical results under a common formalism but also opens the door to new research directions. Future work will investigate finite-blocklength regimes and coding schemes that approach the derived capacity bounds under joint security and robustness constraints. Another promising avenue is the integration of these theoretical insights with machine-learning-based detection models, enabling the development of stegosystems that remain provably secure even against adaptive and data-driven adversaries.

7 Future Work

Several directions remain open for future research. A natural extension is the derivation of finite-blocklength bounds and second-order asymptotics for the ϵ -embedding capacity, which would refine the asymptotic results presented here and provide practical insights into the performance of systems operating with limited block lengths. Another important line of inquiry is the joint characterization of embedding capacity under simultaneous security and robustness constraints, an open problem that lies at the heart of practical design.

Further work should also address the integration of computational notions of security into the information-theoretic framework. Bridging these two paradigms would allow for positive embedding rates under realistic adversarial assumptions, even when perfect statistical security is unattainable. Finally, it is of significant interest to explore coding schemes that approach the derived fundamen-

tal limits, as well as to investigate machine learning-based embedding and detection methods within the proposed theoretical framework. Extending the analysis to more complex and structured cover models, beyond the i.i.d. assumption, remains another challenging but promising research avenue.

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